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Model hybridization & learning rate annealing for skin cancer detection

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Abstract
The increasing frequency of skin tumour across the globe and their timely diagnosis is one of the most promising research directions in the healthcare domain. The most important cause behind the skin cancer mortalities is delayed detection. Early detection followed by adequate treatment may enhance the chances of human survival to a great extent. However, extracting the features from the tumour images for the possible detection of skin cancer is not a trivial task. Numerous deep learning models are extensively employed for the efficient features' extraction for skin cancer detection but literature demonstrate further scope of improvements in various performance measures. In this paper, we propose a hybrid deep Convolutional Neural Network architecture inspired from pretrained architectures for the skin cancer detection by incorporating three major heuristics viz. usage of multiple smaller sized convolutional filters instead of using a single larger filter homogeneously and consistently across the entire model, utilization of skip or residual connections to mitigate the vanishing gradient problem in the deeper model, and learning rate annealing by introducing cyclic learning rate. Experimental results performed on HAM10000 dataset observed an improvement in various performance measures and faster model convergence to a significant extent in comparison with the state-of-the-arts.

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1 Introduction

Cancer is a generic term for a large group of diseases that can affect the human survival to a measurable extent. Cancer is the second leading cause of death globally and about one out of six deaths is due to cancer among all the mortalities caused due to various diseases. Approximately 70% of deaths from cancer occur in low- and middle-income countries and around one third of deaths from cancer are due to the five leading behavioral and dietary risks factors viz. *high body mass index*, *low fruit and vegetable intake*, *lack of physical activity*, *tobacco-*, and *alcohol-consumption*. As per one of the statistics, Cancer is responsible for approximately 7.8 million deaths worldwide as of October 2020 [30]. There are various types of cancers such as skin-, breast-, lung-, bladder-cancer, and many more. Skin cancer is the most common form of cancer, globally accounting for at least 40% of cancerous cases. More than 90% of skin cancers are caused by exposure to ultraviolet radiation of Sun. People with lighter skin are at higher risk due to poor immune system, inappropriate medications, or HIV infections. Skin tumor screening is the most important step for the possible and early identification of skin malignant growth; however, it is not always feasible for dermatologists to check all the patients having skin related issues for skin tumors. In many nations, essential watchfulness is kept through essential consideration facilities before being referred the patient to dermatologists. Subsequently, around 20% patients counsel at essential consideration centers whining skin-related indications [2]. An investigation by Julian et al. revealed that among these 20% patients, 16% patients with dermatological abnormalities had non-harmful skin tumors, 3.3% had actinic keratosis, and 3% had harmful skin tumors [9]. Similar study by Kerr et al. [23] revealed that 11.4% of examined patients had benevolent skin tumors with less than 5% suffering from threatening skin tumors. In spite of the fact that the patients with harmful skin tumors who counsel at essential consideration facilities isn't so high, essential consideration specialists are under a substantial workload to accurately screen patients presenting skin-related side effects and figure out patients to be referred to the dermatologists for further screening. Conclusively, any gadget or automated tool that can precisely give the likelihood of threat by examining a tumor image would really be useful for all i.e. essential consideration specialists, dermatologists, patients, and their relatives.

Tumors emerging from the skin are collectively called as skin-cancers, resulting from an abnormal growth of skin cells or tissues that are capable of invading or spreading to other cells, tissues, or other parts of the body. Broadly, skin-cancers are classified into three categories viz. basal cells, squamous cells, and melanoma. The former two are generally referred as nonmelanoma skin cancers. Basal cell cancer develops slowly and can affect the skin tissues but it is unlikely to spread it in distant areas. However, it is more likely that squamous-cell skin cancers do spread in distant areas. It typically occurs with a scaly top as a rough lump but can also form an ulcer. The most threatful category is melanomas that may affects human survival to a greater extent.

The challenges in the medical image processing includes the lack of annotated data, unavailability of high-resolution images, handling heterogeneity and imbalances in the data, chances of human errors, etc. Manual processing of these data by experts has a limited accuracy and mostly time consuming. Skin related abnormalities are very common in human but, unfortunately, the number of expert dermatologists is fairly low to take all cares of each patients. Furthermore, clinical diagnostic accuracy is also a bit disappointing as lots of morphological features are not even visible through the naked eyes. Moreover, other non-invasive imaging techniques further reduce the diagnostic accuracy, if handled by inexperienced practitioners and their subjective judgment. As the picture is taken through endoscope mostly, a major portion of the picture remains dark. Hence, efficient and meaningful representation of the visual features from these dull/light images for the possible detection of skin tumors with the advances of Convolutional Neural Networks (CNNs) is a challenging task.

Artificial Intelligence (AI) is a branch of computer science that deals with the smart and automated tools or systems capable of performing tasks that typically require human intelligence. AI is an interdisciplinary science offering a paradigm shift in almost every sector of the tech-industries with the advances of machine learning (ML) and deep learning (DL). Healthcare domain is not an exception in leveraging and exploring the benefits of AI. It offers numerous advantages over traditional analytics and clinical decision-making techniques in healthcare domain. Learning algorithms become more precise and accurate as they learn from robust data, allowing medical practitioners to gain unprecedented insights into diagnostics, care processes, treatment variability, and diagnostic outcomes. Deep learning is a branch of AI that mimics the human brain's operations in the presence of huge amount of data and appropriate computational power by employing multiple layers of neural networks. It marks the successes in various domains such as *object detection*, *speech recognition*, *language translation*, and *underlying decision making*. Convolutional Neural Network is a deep learning model, gained huge popularity in computer vision domain. Based on their weight sharing mechanism, sparsity in connections, and translation invariance properties, CNNs are also known as shift invariant or space invariant artificial neural networks (SIANN). The various advancements of the CNNs such as Pretrained CNNs, Residual Connections, and different types of convolutions are being proposed for better features' extraction and classification task.

AI has the potential to decrease dermatologist workloads, eliminates repetitive and routine tasks with enhanced accuracy, and improves access to dermatological care. In this paper, we propose a hybrid deep CNN architecture inspired from Visual geometry group (VGG) and AlexNet architectures for the skin cancer detection. Our model utilizes the strengths of various pretrained CNN architectures to generate a hybrid architecture for enhancing the skin-cancer detection performance. We incorporate three heuristics in the hybrid CNN architecture for improving the model performance and convergence statistics. We employed multiple smaller sized convolutional filters instead of using a single larger filter homogeneously and consistently in the model, resulting lighter model complexity. Hence, the model goes deeper, resulting in vanishing gradient problem in the model training. To address this problem, residual or skip connections are introduced across different pair of consecutive blocks of the model as residual links as seen in ResNet architecture are utilized to tackle any performance limitations caused due to vanishing gradients problem. Lastly, learning rate annealing is introduced by applying cyclic learning rate while training the model. We evaluated the performance of proposed hybrid architecture for multiclass skin cancer classification task on the HAM10000 dataset, divided into seven output categories namely Actinic keratoses, Basal cell carcinoma, Benign Keratosis, Dermatofibroma, Melanoma, Melanocytic nevi, and Vascular lesions.

The proposed hybrid architecture achieves an accuracy of 95.56% approximately, around 1.5% increase in the comparison with the state-of-the-arts. Moreover, we observe a significant improvement in the other performance measures such as sensitivity and specificity, and faster convergence of the model requiring around half the number of epochs for model training in comparison with [14]. A comparison of our proposed model with state-of-the-arts and an analysis of key trends and improvements over [14] has been highlighted in detail in the results section.

The rest of the paper is organized as follows: Section 2 highlights the relevant literature for the skin cancer detection using machine learning and deep learning models. Section 3 presents the major portion of the paper covering dataset and its preprocessing, partitioning the dataset in train, test, and validation set, deep CNN along with pretrained CNN architectures, architectural details, and underlying concepts of the proposed model. Section 11 covers the gradient descent optimizer employed in model training and various performance measures used to evaluate the proposed model. Section 12 presents the results of the experimentations along with the discussions. Conclusions and future research directions are summarized in last section.

2 Related works

The status of latest research in the skin cancer detection with deep learning can be summarized using four different aspects. In literature, binary (cancerous/non-cancerous) as well as multiclass classification (determination of type of cancer) both are extensively experimented for the skin tumor detection and classification, as a first aspect. The second aspect includes the various characteristics of the underlying dataset such as grey scaled or colored images of tumors, the amount of preprocessing, handling imbalance dataset, inadequacy in the input image quality, noise related to annotations, smaller sized dataset, etc. Third aspect is the ensemble of CNNs or other deep models with traditional ML architectures. Generally, CNNs act better as feature extractor rather than a classifier. Therefore, most of the researchers utilized CNN as a feature extractor and other ML classifiers such as Support Vector Machine (SVM), Bayesian are employed for classification. Fourth aspect can be considered as deciding the starting point of model training by introducing a concept called transfer learning. Various pretrained models such as AlexNet, ResNet, VGG are experimented to further fine-tune any proposed architecture for skin cancer detection using deep models.

The advancements of ML/DL models leverage the performance improvements in various tasks from different domains. Skin cancer using Artificial Neural Network (ANN) and ML/DL models has been widely researched in the literature and benchmarking results were obtained on different datasets. Cireşan et al. [12] applied multi-segment profound neural systems for picture classification. Various papers were presented in picture order utilizing profound convolutional neural systems, for example, Chen et al. [11], Wei et al. [19], Li et al. [24], Zeiler et al. [34], and Srdjan et al. [31].

Numerous research works have been published on identifying, segmenting, and classifying skin cancers using various computer vision and deep learning techniques. Using a CNN, Esteva et al. [15] established the skin cancer classification and tested the GoogLeNet' Inception-v3 pre-trained model on a set of 1,29,450 images dataset, handcrafted by the researchers at Stanford. The key feature here was to modify the output classes based on the given task and the model worked well with binary as well as multiclass classifications and was found to be 72.1% accurate and producing a sensitivity of 96%. Furthermore, Hitoshi et al. [31] introduced an automated method for the classification of melanomas. While Anas et al. [5] categorized melanoma into four classes using K-means clustering and employed six different local binary patterns for features' extraction and enhancing the image quality during pre-processing, resulting an average accuracy of 83.33%. In a similar study, Almansour et al. [3] have also implemented deep model for melanoma classification using K-means clustering and SVM.

Subsequently, Abbas et al. [1], and Capdehourat et al. [8] developed binary classifiers over a set of images collected from the internet. They employed CNN for feature extraction and model gained an overall sensitivity of 89.28% and a specificity of 93.75%. On the other hand, Capdehourat et al. adopted AdaBoost C4.5 decision trees to classify skin lesions and used *hair removal algorithms* (HRA) to enhance the quality of input images. In [28], a decision support system using image recognition and neural network algorithm is developed which is based on texture, colour, visual diagnostic attributes, infected area, and degree of damage to the melanoma, with the help of Bayesian and K-Means classifiers coupled with multilayer perceptron and achieved an accuracy, sensitivity and specificity of 73.47%, 76.47%, and 70.21% respectively, for a cluster size of seven.

Melanoma classification model proposed by Blum et al. [7] achieved an accuracy of 87.2%, pertained on the ECOC SVM model, tested on a set of roughly 900 images collected by the University of Tubingen. Kiran et al. [25] created an Android framework for the classification of melanoma with entire workflow built using OpenCV and provided an on-to-go solution for basic diagnosis with an accuracy of 67% and sensitivity of 80.5% for a dataset of images collected from the internet. A multiclass classification using ECOC SVM and deep CNN is developed by Dorj et al. [14]. The approach was to use ECOC SVM with pre-trained AlexNet CNN and classify multiclass data. An average accuracy of 95.1% is reported in this work. Han et al. [17] used a deep CNN to classify the clinical images of 12 skin diseases with a classification accuracy ranging from $95-97\%$.

Vijayalakshmi et al. [32] offers a fully automated dermatological disease recognition method with the help of lesion images, an involvement of a software in contrast to Conventional medical staff-based diagnosis is utilized. The model has been built in three stages – (1) data collection and improvement, (2) architectural design model, and (3) predicting the results. The model with additional Otsu segmentation for pre-processing was tested on the ISIC dataset and measured to be 85% accurate. They have used several AI algorithms such as CNN and SVMs, and combined with image processing software to construct a stronger structure. Hasan et al. [18] proposed a framework for the identification of skin cancer using image recognition and machine learning. Characteristics of damaged skin cells were extracted after segmentation of the dermoscopic images using a novel feature characterization technique. A method focused on deep CNN classifier which was used for stratification of the features' extraction and validated using ISIC dataset, achieved an accuracy of 89.5%. Further, Saket et al. [10] proposed a model for skin cancer detection, and validated using 938 unidentified sample images from the validation set. To assess the MobileNet model's output on unknown images from the validation collection, the micro and weighted averages for precision, recall, and F1-score were utilized. Advancements of the CNNs are also evaluated for skin cancer detection in [26] wherein the benefits of transfer learning and dilated convolutions coupled with the pre-trained InceptionV3 model to classify among seven different skin lesions in the HAM10000 dataset and achieved an overall weighted average accuracy of 89%. The major contribution of various deep learning models especially CNNs in skin cancer detection along with the underlying datasets and results are summarized in Table 1.

3 Materials and methods

This section covers the dataset utilized for the experimentations along with underlying pre-processing, distribution of the dataset into train, test, and validation set. In addition, we present

Table 1 The summarization of existing work for skin cancer detection using deep models

Ref.	Task	Dataset	Model used and Key Contributions	Results
[14]	Skin Cancer Classification	RGB Images collected from the Internet	• Multiclass classification (4 Classes) • ECOC SVM as classifier • CNN as features extractor	Accuracy 95.1% (squamous cell carcinoma) Sensitivity 98.9% (actinic keratosis) Specificity 94.17% (squamous cell carcinoma)
[15]	Skin Cancer Classification	Handcrafted dataset of 129,450 images comprising from 2032 various cancer diseases	• Multiclass classification (2–4 classes based on tasks) • Usage of Google Inception v3 pretrained model • CNN as features extractor	Accuracy 72.1% Sensitivity 96%
[7]	Melanocytic lesions detection	837 melanocytic lesion samples collected from Dermatology Dept., Univ. of Tübingen	• Multiclass classification (4 Classes) • ECOC SVM as classifier • CNN as features extractor	Accuracy 87.2% Sensitivity 92.6% Specificity 87.2%
[28]	Melanoma Classification	98 previously diagnosed dermoscopic images	• Multiclass classification (4 Classes) • Bayesian Classifier, K-Means • Cluster size 7 • Multilayer Perceptron	Accuracy 73.47% Sensitivity 76.47% Specificity 70.21%
[9]	Skin Cancer Classification	HAM10000 Dataset	• Multiclass classification • CNN as Features' extractor • Use of ResNeXt101 pretrained model	Accuracy 92.83% for ensemble model
[8]	Melanoma Detection	635 Images collected from the internet	• Binary classification (2 Classes) • AdaBoost with C4.5 decision trees • Hair removal Algorithm for pre-processing	Sensitivity 90% (malignant melanocytic Levi) Specificity 85% (malignant melanocytic Levi)
[5]	Melanoma Detection	150 images collected by University of Waterloo	• Binary classification (2 Classes) • K-means clustering • Local Binary patterns as features extractor	Accuracy 83.33%
[25]	Skin Lesion Classification	83 RGB images collected from the internet	• Binary Classification • OpenCV kNN Classifier	Accuracy 66.7% (benign melanoma)

Table 1 (continued)

Ref.	Task	Dataset	Model used and Key Contributions	Results
			• OpenCV2.2 for image processing	Sensitivity 80.5% (benign melanoma) Specificity 60.7% (malignant melanoma)
[32]	Melanoma Skin Cancer Detection	ISIC Dataset	• Binary classification • Back Prop, SVM and CNN • Otsu Segmentation for pre-processing	Accuracy 85%
[18]	Skin Cancer Detection	ISIC Dataset	• Binary classification • Back Prop and CNN	Accuracy 85% (malignant lesion) Sensitivity: 84% (malignant lesion)
[17]	Cutaneous Tumour Classification	Asan Dataset, MED-NODE Dataset	• Multiclass classification (12 Classes) • ResNet-152 • CNN+Grad-CAM as features extractor	Accuracy 91% (Average) Sensitivity 86.4% (Average) Specificity 85.5% (Average)
[10]	Skin Cancer Detection	HAM10000 Dataset	• Mobile-Net Model	Accuracy 89% (Weighted average) Recall 83% (Weighted average) F1-score 83% (Weighted average)
[26]	Skin Lesion Classification	HAM10000 Dataset	• Dilated InceptionV3 Model • Transfer learning - ImageNet	Accuracy 89% (Weighted average) Recall 89% (Weighted average) F1-score 89% (Weighted average)

deep CNN including pretrained CNNs, their architectural details, and underlying concepts of the proposed model.

3.1 Dataset, characteristics, and pre-processing

The dataset utilized in our experimentation is HAM10000 which consists of 10,015 dermoscopic images that serves as a classic dataset for the skin cancer detection using any machine learning or deep learning model. Instances of the dataset include a representative collection of all-important diagnostic categories in the realm of pigmented lesions: Actinic keratoses and intraepithelial carcinoma / Bowen's Disease (*akiec*), basal cell carcinoma (*bcc*), benign-keratosis-like lesions such as solar lentigines / seborrheic keratoses (*bkl*), dermatofibroma (*df*), melanoma (*mel*), melanocytic nevi (*nv*), and vascular lesions such as angiomas, angiokeratomas, pyogenic granulomas and haemorrhage (*vasc*). Figure 1 shows the distribution of instances into different categories and several sample lesions images of each category. From this category distribution, we can visualize the inherent imbalance in the dataset classes. Hence, dataset preprocessing is performed to remove the bias among different classes and to normalize the features [20].

The aim of preprocessing is an improvement of the image data that suppresses unwanted distortions and enhances some image features that could be important and beneficial for further processing. Neighboring pixels corresponding to an object in an image should have essentially the same or similar brightness characteristics. Thus, distorted pixel can often be restored as an average value of neighboring pixels. When pre-processing aims to correct some degradation in the image, the nature of a priori information is important such as knowledge about the nature of the degradation, knowledge about the properties of the image acquisition device, conditions under which the image was obtained, etc. The nature of noise (usually its spectral characteristics) is sometimes known. In case of HAM10000 dataset used for model training, the data not carrying any unwanted noise. Therefore, only the size reduction was used as a tool in order to obtain the instances with proper dimension to be provided as input to the proposed model.

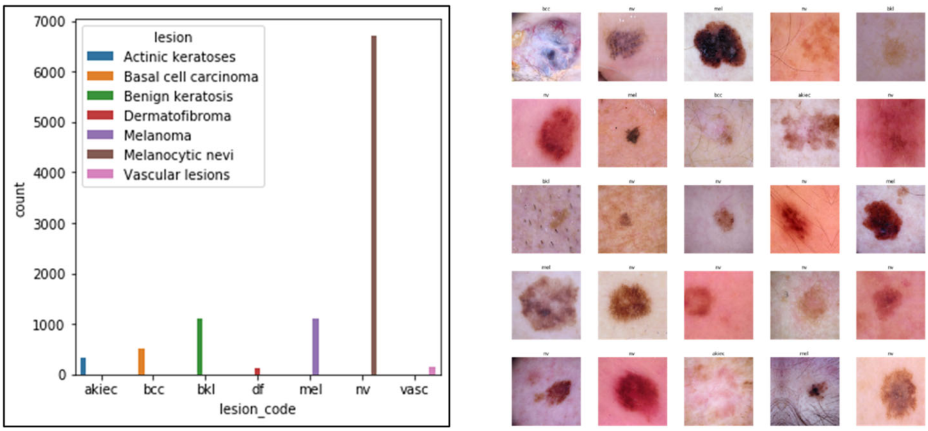
3.2 Data partitioning

We divide the dataset in train, test, and validation set in the proportion of 75%, 5%, and 20% respectively for the model training and evaluation, as illustrated in Table 2.

As the dataset is imbalanced, we also ensure the participation from all the classes or lesion types in each set i.e., training, testing, and validation set, as illustrated in Table 3.

3.3 Convolutional neural networks and pretrained models

Deep learning is emerged in 2006 from machine learning wherein deep neural networks are architectured for the minimization of the loss or error components. It incorporates



(a) (b)

Fig. 1 (**a**) Distribution of instances across different classes and (**b**) several sample lesions of HAM10000

Table 2 Proportion and number of instances in train, test, and development set

Dataset type	% Image portion of the total	Number of images
Training	75	7299
Testing	5	679
Validation	20	2716

representational feature learning and synthesizing features in incremental fashion using multiple layers of neural networks. Convolutional Neural Networks are the important class of deep models and majorly employed for solving computer vision tasks.

CNN can learn the important features automatically for detection and classification. This speciality of CNNs bypass the explicit and labour-intensive handcrafted features engineering process. Thus, by using CNN, complicated image pre-processing for features selection and extraction is no longer necessary. Due to scaling inefficacy in fully connected neural networks, CNN is widely adopted to capture the spatial and contextual information with fewer parameters. While dealing with high-dimensional inputs, it is almost impractical to connect all neurons in a given layer to all neurons in the previous layer. Instead, we connect each neuron to only a part of the previous layer. This key feature of CNN can successfully capture the spatial as well as contextual features information from the single- and high-dimensional input with the help of relevant filters. The generic architecture of the CNN is presented in Fig. 2 wherein extracted features with the help of convolution layers are given as input to the fully connected layers. Finally, *SoftMax* function produces the probability distribution for the output categories. As an example, if the input images is to be categorized in one of the four different classes, the last layer of the deep model would have four nodes. SoftMax function converts each node's output into probabilities of each class such that summation of those probabilities would sum up to one.

Coming to the architectural design of CNNs, convolutional, pooling, and rectified linear unit (ReLU) collectively act as a basic transformation unit converting an input volume to an output volume. The intuitions behind the layer's operation is illustrated using Fig. 3. The convolutional operation is the main part of the convolutional neural network. The spatial extent in the convolution operation is a hyperparameter known as a receptive-field or filter-size. Filter-size that convolves over the input plays an important role in extracting useful features information. Other hyperparameters such as *depth*, *stride*, and *zero-padding* decide the size of the output volume [6]. Similar to the regular neural networks, a dot product is performed between the weights (w) and the spatial input (x) and any non-linear activation function (f) is applied after adding the bias term (b). Pooling is generally applied to an output

Table 3 Participation from different lesion types in train, test, and validation set

Sr. No	Cancer/lesion Type	No. of training images	No. of testing images	No. of validation images	Total
1.	Actinic keratoses	1064	99	396	1559
2.	Basal cell carcinoma	1010	94	376	1480
3.	Benign keratosis	1118	104	416	1638
4.	Dermatofibroma	1010	94	376	1480
5.	Melanoma	1053	98	392	1543
6.	Melanocytic nevi	1000	93	372	1465
7.	Vascular lesions	1043	97	388	1528

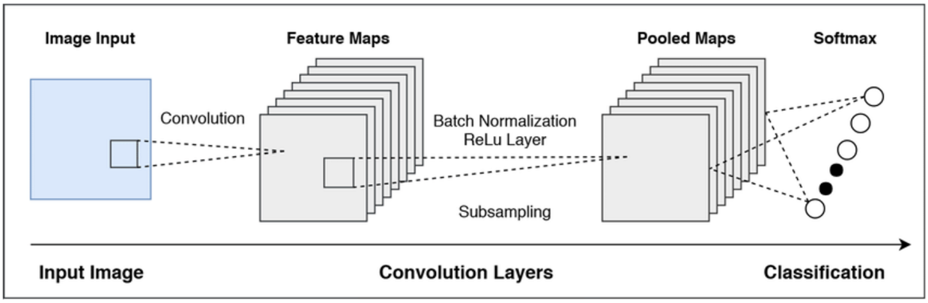


Fig. 2 Generic Architecture of CNN [33]

of the conv-layer and this layer performs a down sampling along the spatial dimensions and is useful in extracting the dominating features. As an example, features such as negation, whether it is in a textual or visual segment in Sarcasm Detection task, must have the highest score in the convolution operation so that it can dominate other scores in a max-pooling operation. There are several types of pooling such as max-pooling, average-pooling, and sum-pooling [4], and each is chosen depending on the application's nature. Max-pooling has found to work better in practice than average pooling for computer vision tasks such as image classification.

In recent years, developed CNNs have been pretrained by research communities so as to perform better for computer vision and other domains' tasks. Several pretrained classic CNN architectures are available and trained on various standard datasets, may provide better starting point for model training for any other specific tasks. We utilized some of them directly or indirectly such as AlexNet, VGG, and ResNet in experimentations pertaining to this research work [27]. These pretrained models are elaborated below in brief as detailed description of these pretrained models is out of the scope of the paper.

3.3.1 AlexNet

The network has a very similar architecture as LeNet by Yann LeCun et al. but it is deeper, with more filters per layer and with stacked convolutional layers. It consists of three different sized filters viz. 11×11, 5×5, and 3×3. The network consists of five convolution layers followed by three fully connected layers, comprising approximately sixty million parameters. The input size of this network is 256×256. Moreover, data augmentation and dropout are the major highlights of the model to achieve generalization and to reduce model overfitting.

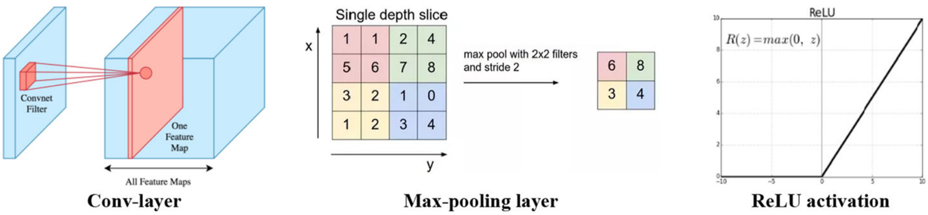


Fig. 3 CNN Operations [13]

3.3.2 VGG

VGG was conceived out of the need to lessen the number of boundaries in the conv-layers and to reduce model training duration. It consists of 16 convolutional layers and used quite too often due to its uniform architectural style. VGG16 has a total of around 138 million parameters. The important point to note here is that all the conv kernels are of size 3 × 3 and max pool kernels are of size 2 × 2 with a stride of two. Though, we are not utilizing this architecture directly in our research work, idea of consistent smaller sized filters was taken from this architecture to improve the model consistency and overall training time.

3.3.3 ResNet

Deep Residual Network is almost similar to the networks which have convolution, pooling, activation and fully-connected layers stacked one over the other. A shortcut connection or skip connection is introduced which turns the network into its counterpart residual version, as illustrated in the Fig. 4. It is one of the most effective architectural methods to handle vanishing gradient problem in deeper networks. The identity shortcuts $f(x\{W\} + x)$ can be directly used when the input and output are of the same dimensions, generally known as identity connection, represented using *solid lines* in the architectures. When input and output differs in dimensions, projection shortcut is used to first match the dimensions followed by performing the task of residual connection, represented using the *dotted lines* in the architectures.

3.4 Proposed model

The proposed architecture is an attempt to improve the existing performance measures and to minimize the convergence time of deep learning model presented for skin cancer detection in [14]. Here, we introduced and incorporated uniformities in convolution filters, usage of relatively smaller filters, mitigating the vanishing gradient problems using skip connections

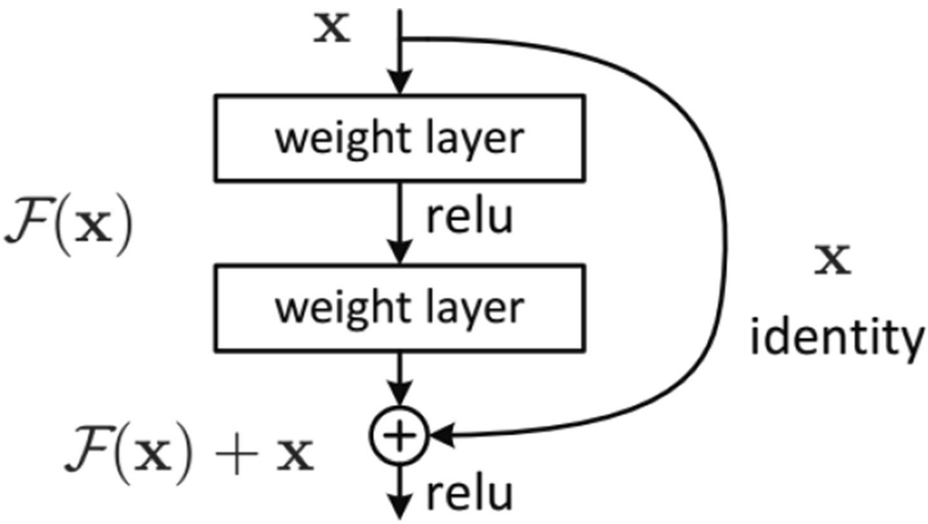


Fig. 4 Skip connection in the residual network module

(identity- and projection-shortcuts), and applying learning rate annealing using cyclic learning rate. Uniformity in the convolution filters and usage of relatively smaller-sized filters are combined, and termed as hybridization of pretrained models wherein large-sized convolutional filters of AlexNet are implemented using multiple smaller sized filters along with the usage of residual connections. The idea is to utilize the AlexNet as a pretrained architecture and replacing its larger filters with multiple smaller sized filters all across the deep model to reduce the model parametric complexity for faster model convergence and improved performance, as architectured in the VGG. The depth of the transformed model increases, resulting in vanishing gradient problem in the model training. To address this problem, residual or skip connections are introduced across different pair of consecutive blocks of the model as appeared in the ResNet architecture. Lastly, learning rate annealing is applied by employing cyclic learning rate while training the model. The flow of the process of the proposed model can be summarized with the help of Fig. 5.

Use of consistent multiple smaller-sized filters in the place of a single large-sized filter can certainly improves the parameter space that plays a vital role in the convergence of any deep model. In this experimentation, 3 × 3 and 1 × 1 filters are selected as the basic building blocks for the model, used consistently across the entire deep architecture. Each of the larger convolution filters are then split into a combination of multiple smaller sized filters. The idea behind having fixed size kernels is that all the variable size convolutional kernels used in AlexNet (11 × 11, 5 × 5, 3 × 3) can be replicated by making the use of multiple 3 × 3 kernels as building blocks. The replication is in terms of the receptive field covered by the kernels. For example, single 5 × 5 convolution filter can be implemented by using two 3 × 3 filters, leveraging lesser model parameters and henceforth model complexity, and the same is illustrated in Fig. 6. This can reduce the trainable parameters up to 50%, improving model convergence, and produces robustness to the model towards overfitting.

Experiments have proven that the accuracy actually decreases by adding more layers to the network. This problem is due to vanishing gradient i.e. as we make the CNN deeper, the derivative when back-propagating to the initial layers becomes almost insignificant and some of the parameters might be under-represented. The solution to this problem was hidden in the skip or residual connections of ResNet model. Here, we are able to address these problems using two types of connections viz. projection- and identity-shortcut.

Identity connections are between every two convolution layers. Rather than taking in the mapping from $x \rightarrow f(x)$, the network takes in the mapping from $x \rightarrow f(x) + g(x)$. At the point when the element of the input x and yield $f(x)$ is equivalent, the capacity $g(x) = x$ is an identity mapping and the alternate way association is called identity connection. The identical mapping is found out by focusing weights in the intermediate layer during preparation, since it's simpler

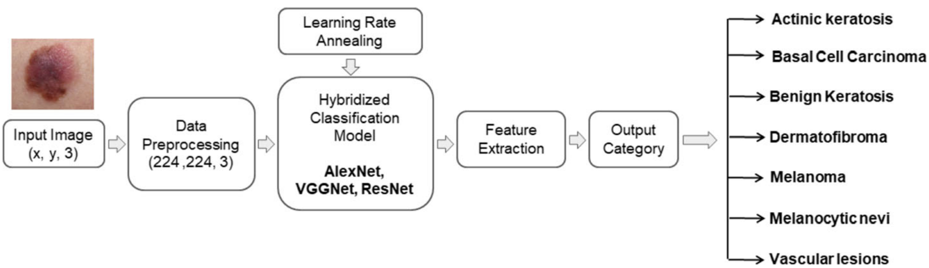


Fig. 5 Flow of the proposed model

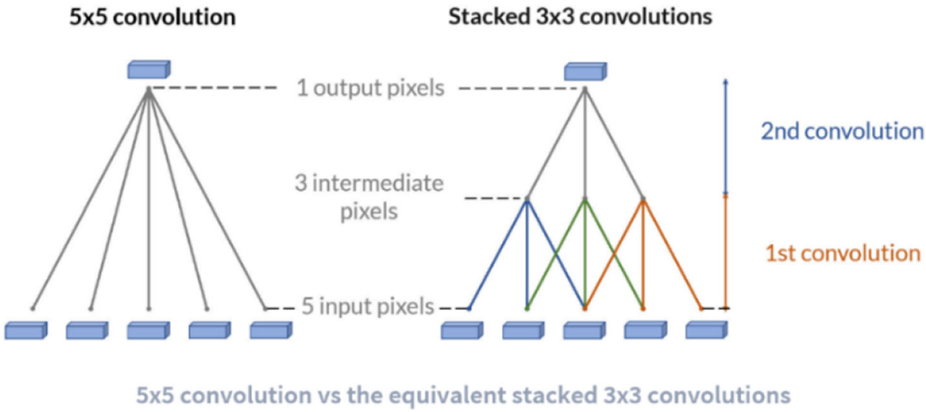


Fig. 6 Replacement of a larger filters by multiple smaller sized filters

to zero out the load than push them to one. These additional connections between nodes of different layers that usually skip one or more layers of nonlinear processing have actually made training of deeper networks possible as it rightly takes out the singularities caused by the non-identifiability of the model.

The skip connections allow information to flow in the skip layers, so, in the forward pass, information from layer l_1 can directly be fed into layer $1 + l_1 + t$ for $t \geq 1$, where t is the number of layers skipped by the skip connections. During the backward pass, the gradients can also flow unchanged from layer $1 + l_1 + t$ to layer l_1. As the gradients remain unchanged across a block of conv-layers, we mitigate the vanishing gradient problem to a considerable extent.

The learning rate is a hyperparameter that controls the change in the magnitude of the model parameters in response to the estimated error in the gradient descent process. Choosing the learning rate is a bit challenging task as a small learning rate may result in a long training process. However, high learning rate may result in learning a sub-optimal set of weights too fast or an unstable training process, and may not be able to converge the model at all.

Apart from this, other problems encountered due to selection of poor learning rate include network getting stuck at either at the saddle points or the local minima. Low learning rate might not be sufficient to break out of the area and descent into areas of lower loss landscape or our model may be highly sensitive to the initial learning rate. These problems can be easily rectified with the help of cyclic learning rates (CLR) as it provides the freedom in selecting out initial learning rate choices and also break out of any saddle points if necessary. Cyclic learning rates allows the learning rate to oscillate back and forth between the lower and upper bounds while model training, slowly increasing and decreasing the learning rate after a predefined number of epochs. The simplest and perhaps most used adaptation of learning rate during training are techniques that reduce the learning rate over time. The intuition behind this approach is that we'd like to traverse quickly from the initial parameters to a range of "good" parameter values but then we'd like a learning rate small enough that we can explore the deeper, but narrower parts of the loss function. These have the benefit of making large changes at the beginning of the training procedure with larger learning rate and decreasing the learning rate such that a smaller rate and therefore smaller updates are made to the parameters later in the training procedure. This has the effect of quickly learning good weights early and fine tuning them later.

Two factors needs to be defined while using a CLR: (1) The bounds between which the learning rate will vary i.e. base learning rate (*base_lr*) and maximum learning rate (*max_lr*), and (2) The step size i.e. how many epochs are required for the learning rate to reach from one bound to the other. In CLR, the learning rate varies between the lower and higher thresholds. The logic is that intermittent higher learning rates in each epoch helps us to break out of any saddle points or local minima, if it is encountered. If the saddle point happens to be an elaborate plateau, lower learning rates will possibly never produce enough gradient to get out of it, making it impossible to mitigate the loss further. There is a simple way to estimate reasonable minimum and maximum boundary values with one training run of the network for a few epochs. It is known as "*LR range test*"; run the model for several epochs while letting the learning rate increases linearly between low and high learning rate values. This test is enormously valuable whenever you are experimenting on a new architecture or dataset.

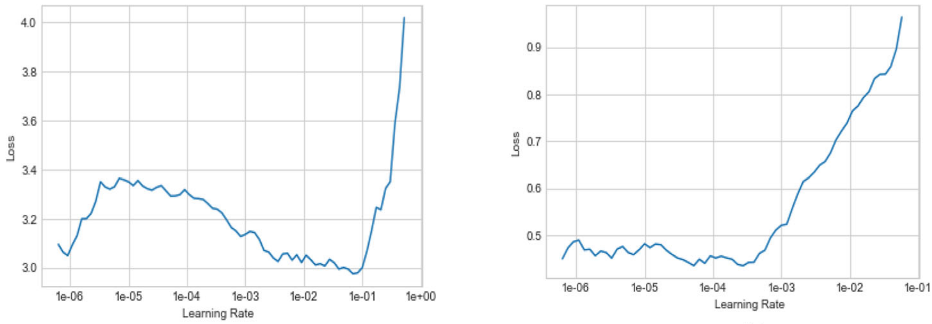
In our experimentation, the model was made to train for 10 epochs and the learning rates were gradually tweaked and the learning loss for each value was noted. These values were later used to plot the LR range graph and the point with the lowest learning loss was selected as the *max_lr* value, as illustrated in the Fig. 7(a). Further, the graph was used to identify regions of steep descent in learning rate and the region with maximum descent is best suited for selecting the base learning rate. The region between 10e-4 and 10e-3 in Fig. 7(b) has the maximum decline in slope and the base LR must lie in this zone. The model was again run for a few epochs with the boundaries in mind to figure out an optimum value in the given range. The optimum value for *base_lr* and *max_lr* were found to be around 0.025 and 10e-1 respectively.

With the three aforementioned improvisations viz. model hybridization, residual connections, and learning rate annealing applied to the model, the detailed architectural aspects and parametric setup is summarized as follows and the same is presented in Fig. 8.

- Input layer (IL): The size of the skin malignancy's crude RGB image is $227 \times 227 \times 3$.
- Layer 1 (CL): This layer is the first feature extraction layer of our network. 64 filters of size 7×7 each employed in this layer, and padding of 1 is used to generate the feature maps with stride 2. The output of this convolutional layer is $(229–7) / 2 + 1 = 112$. Then output tensor shape will become $112 \times 112 \times 64$.
- Layer 2 (MPL): The Max-Pooling layer, which uses a 3×3 filter, and stride 2. The output size of this layer is calculated by $(112–3) / 2 + 1 = 56$. Depth is same as layer 1. The output tensor shape would be $56 \times 56 \times 64$.

Block 1 This Block termed as Conv_2x has 3 different convolutional layers for feature extraction. Each of these layers differ in the number of filters and filter-size. This block is repeated 3 times and comprises of the next 9 layers as mentioned below.

- Layer 3, 6, and 9 (CL): This is the second convolution layer with 64 filters. The filter size is 1×1, stride is 2 for layer 3 and stride is 1 for layer 6 and 9, and zero padding is applied. Because of zero padding, the original size is restored. The output tensor shape would be $56 \times 56 \times 64$.
- Layer 4, 7, and 10 (CL): The third convolution layer with 64 filters for features' extraction. The filter size is 3×3, stride is 1 and depth is 64.
- Layer 5, 8, and 11 (CL): The fourth convolution layer is again a 1×1 convolutional layer with stride 1 and zero padding. This layer has 256 filters and resultant tensor shape will remains unchanged.
- The output size of Block 1 after all 3 iterations is observed to be 56×56.



(a) (b)

Fig. 7 (**a**) Deciding the max learning rate for the model, (**b**) Deciding the base learning rate for the model

- Layer 12 (BN): A normalization layer helps in mitigating the internal covariate shift and overfitting and proven to be a better alternative of dropout regularization.

Block 2 This Block termed as Conv_3x has 3 different convolutional layers. Each of these layers have filter sizes similar to block 1 but the depth has doubled for each of the layers. This block is repeated 4 times and comprises of the next 12 layers as mentioned below.

- Layer 13, 16, 19, and 22 (CL): This is the first convolution layer of conv_3x with 128 filters. The filter size is 1 × 1, stride is 2 for layer 13 and stride is 1 for the rest, and zero padding is applied.
- Layer 14, 17, 20, and 23 (CL): The next convolution layer with 128 filters for features' extraction. The filter size is 3 × 3, stride is 1 and depth is 128.
- Layer 15, 18, 21, and 24 (CL): The last convolution layer of conv_3x is again a 1 × 1 convolutional layer with stride 1 and zero padding. This layer has 512 filters and resultant tensor shape will remains unchanged.
- The output size of Block 2 after all 4 iterations is 28 × 28.
- Layer 25 (BN): Similar to Layer #12.

Block 3 This Block referred as Conv_4x has 3 different convolutional layers. Each of these layers have filter sizes similar to block 2 but the depth has doubled for each of the layers. This block is repeated 6 times and comprises of the next 18 layers as mentioned below.

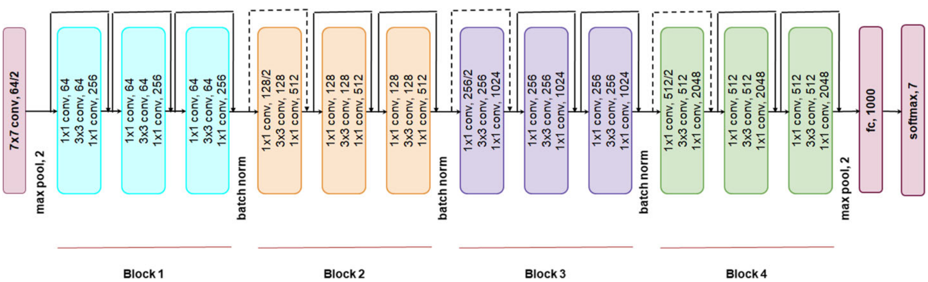


Fig. 8 Architecture of proposed model

- Layer 26, 29, 32, 35, 38, and 41 (CL): This is the first convolution layer of block_3 with 256 filters. The filter size is 1 × 1, stride is 1 for all the layers except layer 26 for which the stride is double, and zero padding size is applied.
- Layer 27, 30, 33, 36, 39, and 42 (CL): The next convolution layer with 256 filters for features' extraction. The filter size is 3 × 3, stride is 1 and depth is 256.
- Layer 28, 31, 34, 37, 40, and 43 (CL): The last convolution layer of batch_3 is again a 1 × 1 convolutional layer with stride 1 and zero padding. This layer has 1024 filters and resultant tensor shape will remains unchanged.
- The output size of Block 3 after all 6 iterations is 14 × 14.
- Layer 44 (BN): Similar to Layer #12.

Block 4 This Block Conv_5x has 3 convolutional layers. Each of these layers have filter sizes similar to block 3 but the depth has doubled for each of these layers. This block is repeated 3 times and comprises of the next 9 layers as mentioned below.

- Layer 45, 48, and 51 (CL): This is the first convolution layer of conv_5x with 512 filters. The filter size is 1 × 1, stride is 1 for layers 48, 51 and 2 for layer 45, and zero padding size is applied.
- Layer 46, 49, and 52 (CL): The next convolution layer with 512 filters for convolution. The filter size is 3 × 3, stride is 1and depth is 512.
- Layer 47, 50, and 53 (CL): The last convolution layer of conv_5x is again a 1 × 1 convolutional layer with stride 1 and zero padding. This layer has 2048 filters and resultant tensor shape will remains unchanged.
- The output size of Block 4 after all 3 iterations is 7 × 7.
- Layer 54 (APL): The average pooling layer uses a 7 × 7 filter and passes its output to a fully connected layer.
- Layer 55 (FCL): Fully connected layer with 1000 neurons. In this layer, each of the 1 × 1 × 2048 = 2048 activations are connected to the 1000 neurons in the fully connected manner.
- Layer 56 (SMX): The last layer of the model architecture is a SoftMax layer with 7 output classes. It gets inputs from the 1000 neurons of the previous layer and uses a probabilistic approach to generate class probabilities for each of the 7 classes.

4 Process optimizer and evaluation metrics

We train the model using the gradient descent approach that internally uses the backpropagation algorithm. For faster convergence of gradient descent, we employ Adam's algorithm as a gradient descent optimizer that internally based on two important concepts viz. momentum and RMSProp. Moreover, it can be seen from the Fig. 8 that Batch Normalization is an integral component of the proposed architecture. Batch-norm standardizes the inputs to the layer for each mini-batch. This has the effect of stabilizing the learning process and significantly reducing the amount of training cycles required to train deep networks. In order to improve the stability of the neural network, it normalizes each layer's activations by subtracting the mini-batch mean from it and dividing it by mini-batch standard deviation. It has been shown to have several benefits as follows:

1. *Networks train faster*: Although, each training iteration would be slower due to extra standardization measurements during the forward pass and introduction of additional hyperparameters in model training, it get converged much quicker, henceforth, cumulative training is faster.
2. *Allows higher learning rates*: Gradient descent typically requires a limited learning threshold for the network to converge. When the network goes deeper, the gradients get smaller during back propagation, requiring even further iterations. Batch normalisation allows utilization of much higher learning rate, improving the model convergence speed.
3. *Makes weights easier to initialise*: Weight initialisation can be difficult, especially when creating deeper networks. Batch normalisation helps in reducing the sensitivity of the model to the initial setting up of the parametric space.
4. *Makes activation functions viable*: Activation mechanisms may not be functioning well in some cases. Sigmoid loses its gradient easily, which means they can't be used in deeper network, and ReLU often die during model training (stop learning completely), so we need to be cautious about the set of values fed into different activation functions. Batch normalization governs the values more viable to feed into these activation function.
5. *Provides regularisation*: Batch normalization brings a little noise to the network, and in certain situations (e.g., Inception modules) it has been seen to perform similar as dropout regularization. Batch normalization can be considered as a bit of extra regularization, helping us to reduce dropouts you could introduced to the network.

Instead of leading the search using just the gradient of the current step, momentum also accumulates the gradient of the previous steps to decide the optimal path. Momentum is also referred as a technique which reduces oscillations in our global minima search in the loss curve. In practice, the coefficient of momentum is initialized at 0.5, and gradually annealed to 0.9 over multiple epochs. RMSProp tries to reduce the oscillations but in a different way than momentum. RMSProp also takes away the need to adjust learning rate and often does it automatically and it's implicitly performs simulated annealing. Adam's optimizer is a combination of both aforementioned techniques i.e. Momentum and RMSProp, illustrated with the help of following equations set. Generally, the hyperparameters utilized in momentum and RMSProp are taken as 0.9 and 0.99 respectively. This helps us in further combatting the problem of pathological curvature and speeding up the global minima search at the same time. $\beta 1$ and $\beta 2$ are the hyperparameters related to momentum and RMSProp respectively. *dw* and *db* are the partial derivatives of the cost function with respective to parameters weights and biases.

In iteration t:

Momentum	**RMSProp**	
$Vdw = \beta 1 Vdw + (1-\beta 1)dw$	$Sdw = \beta 2 Sdw + (1-\beta 2)dw^2$	(1)
$Vdb = \beta 1 Vdb + (1-\beta 1)db$	$Sdb = \beta 2 Sdb + (1-\beta 2)db^2$	(2)
$Vdw^{corrected} = \frac{Vdw}{1-\beta 1^t}$	$Sdw^{corrected} = \frac{Sdw}{1-\beta 2^t}$	(3)
$Vdb^{corrected} = \frac{Vdb}{1-\beta 1^t}$	$Sdb^{corrected} = \frac{Sdb}{1-\beta 2^t}$	(4)

Updating the parameters

$$W = W - \alpha \frac{Vdw^{corrected}}{\sqrt{Sdw^{corrected}}} \quad (5)$$

$$W = W - \alpha \frac{Vdw^{corrected}}{\sqrt{Sdw^{corrected}}} \quad (6)$$

The different performance measures are used for evaluating the various deep learning models based on the nature of the problem. Accuracy, precision, recall, and F1-score are the most important performance measures in case of classification tasks. However, Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) are the performance measures, generally adopted in case of regression problems. As we are focusing on multiclass classification in the detection of skin cancer, we precisely define the performance measures used in our experimentation to evaluate the proposed model. Though, we define the formulas for binary classification, it can be extended for multiclass classification in the similar fashion. The aforementioned performance measures can be computed with the help of confusion matrix, a metric demonstrating the true and false prediction by the model against the ground truth labelling. In case of confusion matrix for binary classification, True Positives (TP) and True Negatives (TN) are the correct predictions. False Positives (FP) and False Negatives (FN) are the positive and negative predictions respectively by the model, contrary to their ground truth labelling.

Accuracy is a ratio of correct classification to the total number of instances, as indicated using Eq. 7. It does not take specific misclassification into account. Precision can be defined as how many samples are TP among all the samples predicted as positives and presented by Eq. 8. However, to take care of FN, Recall is introduced and can be defined as the ratio of TP to the Actual Positives samples. It is presented using Eq. 9, also termed as sensitivity or true positive rate. F1-score is the most important performance measure that takes care of both Precision as well as Recall and can be defined as a harmonic mean of both of these, and expressed by Eq. 10. Specificity is the ratio between how much instances were truly predicted as negative to how much instances were predicted as negative. It is mostly calculated when the probability of occurrence of negatives is higher, and represented in Eq. 11.

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} \tag{7}$$

$$Precision = \frac{TP}{TP + FP} \tag{8}$$

$$Recall/Sensitivity = \frac{TP}{(TP + FN)} \tag{9}$$

$$F1 - score = 2\left(\frac{Precision*Recall}{Precision + Recall}\right) \tag{10}$$

$$Specificity = \frac{TN}{TN + FN} \tag{11}$$

A receiver operating characteristic (ROC) curve, is a graphical plot that illustrates the diagnostic ability of a binary classifier system as its discrimination threshold is varied. The

ROC curve is created by plotting the true positive rate (TPR) against the false positive rate (FPR) at various threshold settings. ROC is a probability curve and Area under curve (AUC) represent degree or measure of separability. It tells how much model is capable of distinguishing among the classes.

5 Results and discussions

We mainly focus on the task of classifying skin cancer images using deep convolutional neural network by employing majorly three heuristics viz. hybrid architecture, residual connections, and learning rate annealing in the direction of improving both performance measures and model convergence statistics. This section presents the detailed results and discussions for the multiclass skin cancer classification using our proposed hybrid architecture on the HAM10000 dataset. We present sample instances of skin tumour lesions used in training and testing of the model (Fig. 9), heatmaps generated by initial and later convolutional layers, category/class wise performance measures of the proposed model, and comparative illustration of our results with [14] and other state-of-the-arts. The hardware used for the experimentation is Intel quad-core i5 8th Gen with 1.60 GHz, 16.00GB Dual-Channel DDR4 @ 665 MHz, NVIDIA Tesla-K80 GPU with 4992 cores. TensorFlow framework along with Python3 and Jupyter Notebooks are utilized for the implementation of the proposed hybrid CNN model. Various performance measures such as precision, recall and F1-score [21], are utilized to evaluate the proposed model. Figure 9 demonstrates few skin malignant samples used in training and testing of the model.

Multiple layered CNN is employed for the skin cancer detection task wherein initial convolutional layers are responsible for extracting the low-level features. However, later layers are responsible for extracting the high-level features. Figure 10(a) shows the result of applying filters on the original input using initial convolutional layer. In order to proceed, the CNN model uses the heatmaps to make prediction, generated by the last convolutional layer, and depicted in Fig. 10(b).

Table 4 presents the class-wise performance of the proposed hybrid model. The maximum accuracy of 99.26% is attained for vascular lesions whereas maximum similarity is observed in the classes melanoma and melanocytic nevi which resulted in lesser accuracy of 87.78% in predicting the melanomas. Least sensitivity/recall of 52.69% is achieved for class melanocytic nevi, although, higher specificity proved that it could rule out all the false positives.

Figure 11 represents the confusion matrix for the improvised model for the various classes for skin cancer detection which shows the summary of the class wise prediction against actual class labelling. We can notice that the highlighted classes melanoma and melanocytic nevi impose similarities which results in indecision in differentiating these two types.

Figure 12 shows the comparative analysis of the proposed model with [14]. Training and testing accuracies for HAM10000 are plotted in the first two graphs. There are two aspects targeted in the experimentation i.e. faster model convergence and enhancing various performance measures. We observe the significant increase in both training and testing accuracy with lesser epochs requirement as compared to [14]. Third graph demonstrates the time taken by the epochs in the model training. Though, time taken per epoch in the improvised model is get increased, the number of epochs is reduced to reach to the same or better convergence point. As the number of layers is increased in the improvised model, more parameters are trained, and can be seen in the fourth graph.

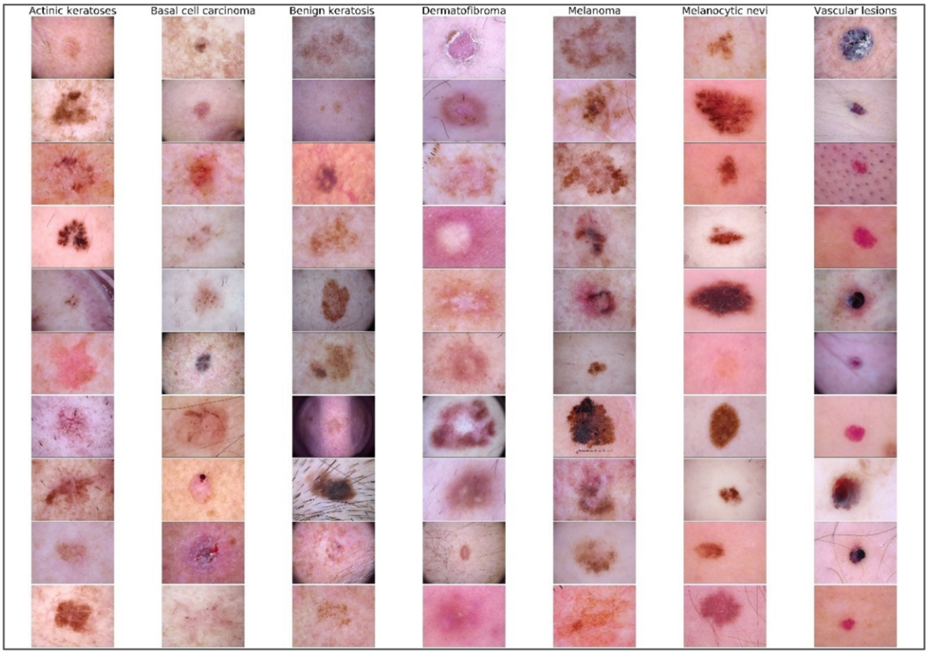
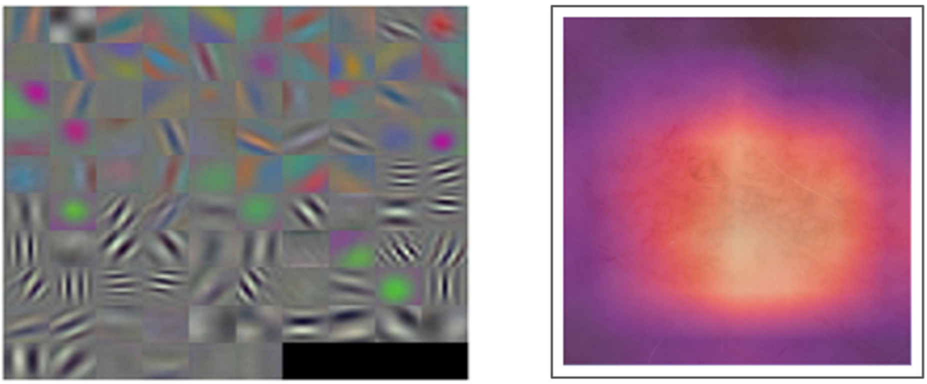


Fig. 9 Sample skin-lesion classes used in training and testing of the hybrid model

Roughly, model is able to gain around 87% accuracy on the training set over the initial 76% in [14] which is approximately a 10% rise in the accuracy. The validation loss is dropped from 0.67 to 0.34 in the improvised model, which is almost a reduction to half. Moreover, all these results are obtained in around 13 epochs which is also half the number of epochs required in [14] (i.e. 25 epochs). It is clearly evident that the proposed hybrid model demonstrates better on both the aspects i.e. performance measures and model convergence time.

A comparative analysis is also performed between performance of proposed hybrid model with the state-of-the-arts and the same is presented in Table 5. By using proposed model, seven kinds of cancers i.e. Actinic keratoses and intraepithelial carcinoma / Bowen's Disease (*akiec*),



(a) (b)

Fig. 10 (**a**) Heatmaps generated through initial and (**b**) later convolutional layers

Table 4 Class-wise performance of proposed model

Cancer Type	Test Images	Precision	Recall	F1-Score	Accuracy	Specificity
Actinic keratoses	99	0.646	0.646	0.723	52.78	97.59
Basal cell carcinoma	94	0.914	0.787	0.846	96.02	98.8
Benign keratosis	104	0.738	0.732	0.734	91.9	95.3
Dermatofibroma	94	0.949	0.787	0.86	96.47	99.32
Melanoma	98	0.542	0.99	0.7	87.78	85.89
Melanocytic nevi	93	0.721	0.527	0.609	90.72	96.76
Vascular lesions	97	1.0	0.948	0.974	99.26	100.0

basal cell carcinoma (*bcc*), benign-keratosis-like lesions (solar lentigines / seborrheic keratoses, *bkl*), dermatofibroma (*df*), melanoma (*mel*), melanocytic nevi (*nv*) and vascular lesions (angiomas, angiokeratomas, pyogenic granulomas and haemorrhage, *vasc*) are classified in our study. As our skin cancer detection classifies in seven output categories, however, most of the

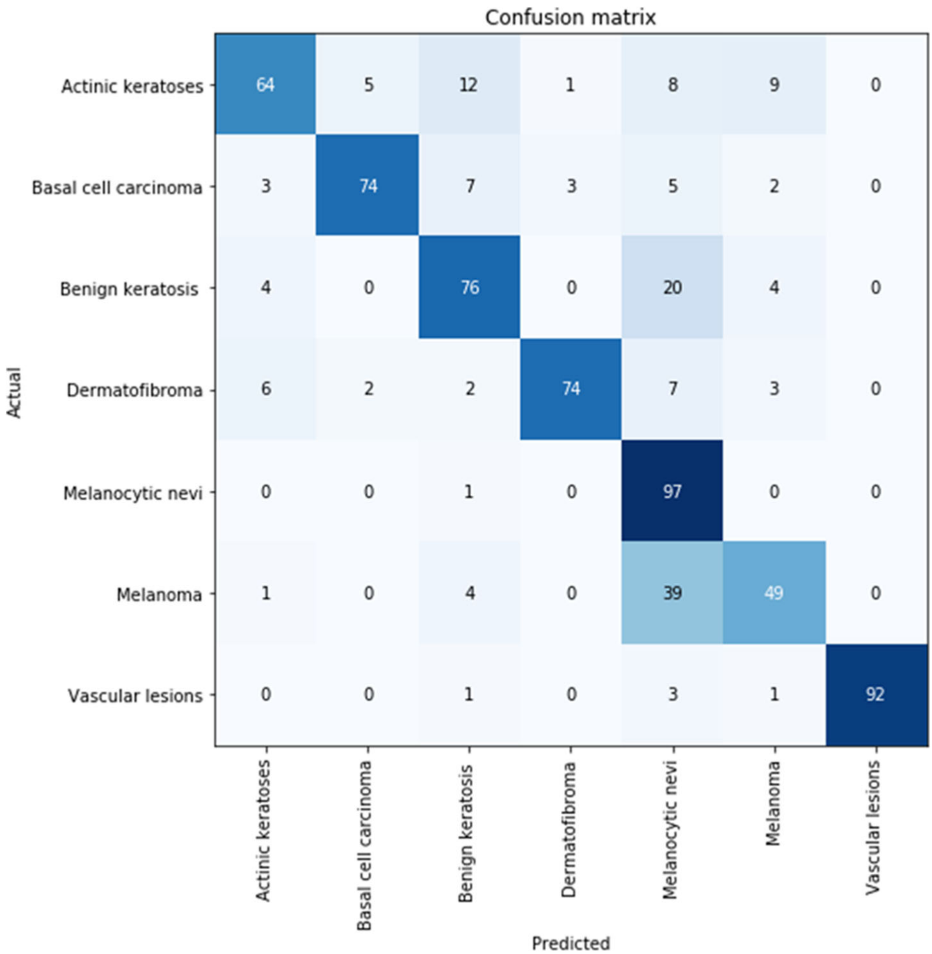


Fig. 11 Confusion matrix for the proposed model in skin cancer detection

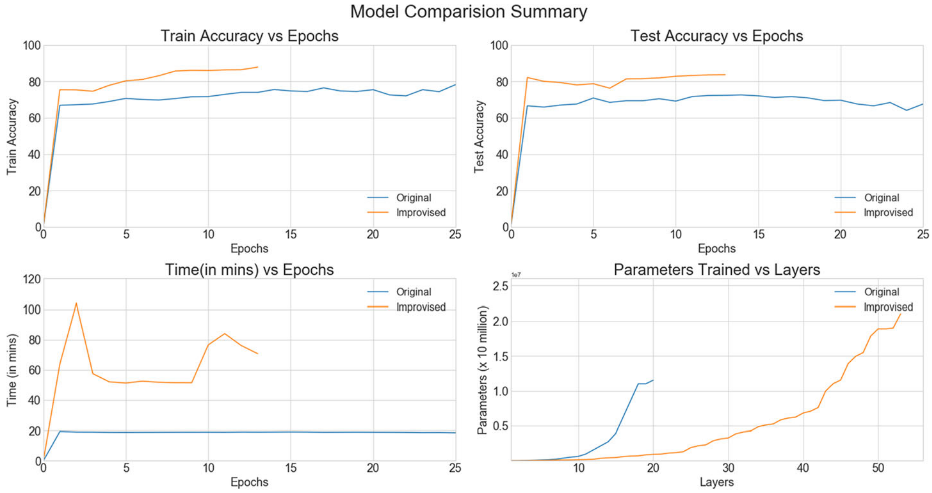


Fig. 12 Comparison of proposed model with [14]

similar studies focus on binary or four class classification. The melanoma is classified by different researchers and actinic keratoses, basal cell carcinoma, squamous cell carcinoma, and melanoma are classified in Dorj et al. [14].

The proposed model's performance is significantly higher as compared to existing models. Moreover, increase in the overall accuracy is also reported. The changes in the sensitivity are not very significant but the specificity value has considerably shot up above the Dorj et al.'s [14] model. As high sensitivity and high specificity play an important role in making predictions in the medical domain, the proposed study will be helpful to further improvise on the predictions.

Table 5 Comparative analysis of the proposed hybrid model with existing works for skin cancer detection

Ref.	% Accuracy	%Sensitivity	% Specificity
Esteva et al. [15]	72.1	96	-
Hitoshi et al. [31]	-	81.1	92.1
Anas et al. [5]	83.33	-	-
Almansour et al. [3]	90.32	93.37	85.84
Abbas et al. [1]	-	89.28	93.75
Capdehourat et al. [8]	-	90	77
Giotis et al. [24]	81	-	-
Ruiz et al. [28]	-	78.43	97.87
Isasi et al. [34]	85	-	-
Blum et al. [7]	87.2	92.6	87.2
Kiran et al. [25]	66.7	60.7	80.5
Dorj et al. [14]	94.2	97.83	90.74
Saket et al. [10]	89	83	83
Ratul et al. [26]	89	89	89
Saket et al. [9]	93.20	88	88
Proposed model	95.56	97.98	96.23

6 Conclusions and future works

We presented the skin cancer detection using hybrid deep CNN model on HAM10000 dataset. The performance measures improvement and faster model convergence are the main objectives while designing the architecture by introducing three heuristics viz. model consistency through multiple sized convolutional filters, residual connections to mitigate the vanishing gradient problem, and learning rate annealing through cyclic learning rate. Moreover, all the seven types of skin cancers (*akiec, bcc, bkl, df, mel, nv*, and *vasc*) are classified using proposed deep convolutional neural model. We conclude from the experimental results that the deeper and consistent model perform better, if suitable heuristics are adopted such as residual or skip connections, learning rate annealing, and other hyperparameters tuning should also be applied for faster convergence of the model. Maximum values of accuracy, precision, and F1-score are recorded as 99.29%, 100%, and 97.4% respectively for vascular lesions. On the downside, 52.78% (actinic keratoses), 54.2% (melanoma) and 60.9% (melanocytic levi) are the least recorded values for accuracy, precision, and F1-score respectively.

The future work will focus on developing model which initially segregates malignant and non-malignant skin-lesions followed by identifying the class of skin cancer, if found to be malignant in the previous segregation. This is known as multilevel skin cancer classification, will be useful in classifying the skin cancers, if the dataset may or may not contain malignancy. In addition to the current research for skin cancer diagnosis, future studies could rely on the use of patient-specific features such as chromosomes, age, and colour. These additional features may be useful in the development of customised and accurate computer-aided system for skin cancer diagnosis. We also hope to expand the development of this model into mobile and web applications which makes it user-friendly and accessible to the masses. This shall help in early detection of the malignancy and hence increase the available time for initial treatment, if necessary.

References

1. Abbas Q, Celebi ME, Serrano C, García IF, Ma G (2013) Pattern classification of dermoscopy images: A perceptually uniform model. Pattern Recogn 46(1):86–97
2. Allwein EL, Schapire RE, Singer Y (2000) Reducing multiclass to binary: A unifying approach for margin classifiers. J Mach Learn Res 1(Dec):113–141
3. Almansour E, Jaffar MA (2016) Classification of Dermoscopic skin cancer images using color and hybrid texture features. Int J Comput Sci Netw Secur 16(4):135–139
4. An Intuitive Explanation of Convolutional Neural Networks. https://ujjwalkarn.me/2016/08/11/intuitive-explanation-convnets/. Accessed 18 Nov 2020
5. Anas M, Gupta K, Ahmad S (2017) Skin cancer classification using K-means clustering. Int J Tech Res Appl 5(1):62–65
6. Bengio Y, Courville AC, Vincent P (2012) Unsupervised feature learning and deep learning: A review and new perspectives. CoRR, abs/1206.5538, 1
7. Blum A, Luedtke H, Ellwanger U, Schwabe R, Rassner G, Garbe C (2004) Digital image analysis for diagnosis of cutaneous melanoma. Development of a highly effective computer algorithm based on analysis of 837 melanocytic lesions. Br J Dermatol 151(5):1029–1038
8. Capdehourat G, Corez A, Bazzano A, Alonso R, Musé P (2011) Toward a combined tool to assist dermatologists in melanoma detection from dermoscopic images of pigmented skin lesions. Pattern Recognit Lett 32(16):2187–2196
9. Chaturvedi SS, Tembhurne JV, Diwan T (2020) A multi-class skin Cancer classification using deep convolutional neural networks. Multimed Tools Appl 79(39):28477–28498

10. Chaturvedi SS, Gupta K, Prasad PS (2020) Skin lesion analyser: an efficient seven-way multiclass skin cancer classification using MobileNet. In: International Conference on Advanced Machine Learning Technologies and Applications. Springer, Singapore, pp 165–176
11. Chen Y, Lin Z, Zhao X, Wang G, Gu Y (2014) Deep learning-based classification of hyperspectral data. IEEE J Sel Top Appl Earth Observ Remote Sens 7(6):2094–2107
12. Ciregan D, Meier U, Schmidhuber J (2012) Multi-column deep neural networks for image classification. In: 2012 IEEE conference on computer vision and pattern recognition, pp 3642–3649
13. Convolutional Neural Networks (CNNs / ConvNets) http://cs231n.github.io/convolutional-networks/. Accessed 22 Nov 2020
14. Dorj UO, Lee KK, Choi JY, Lee M (2018) The skin cancer classification using deep convolutional neural network. Multimed Tools Appl 77(8):9909–9924
15. Esteva A, Kuprel B, Novoa RA, Ko J, Swetter SM, Blau HM, Thrun S (2017) Dermatologist-level classification of skin cancer with deep neural networks. Nature 542(7639):115–118
16. Fujisawa Y, Inoue S, Nakamura Y (2019) The possibility of deep learning-based, computer-aided skin tumor classifiers. Front Med 6:191
17. Han SS, Kim MS, Lim W, Park GH, Park I, Chang SE (2018) Classification of the clinical images for benign and malignant cutaneous tumors using a deep learning algorithm. J Invest Dermatol 138(7):1529–1538
18. Hasan M, Barman SD, Islam S, Reza AW (2019) Skin cancer detection using convolutional neural network. In: Proceedings of the 5th International Conference on Computing and Artificial Intelligence, pp 254–258
19. Hu W, Huang Y, Wei L, Zhang F, Li H (2015) Deep convolutional neural networks for hyperspectral image classification. J Sens 2015
20. Iwendi C, Khan S, Anajemba JH, Bashir AK, Noor F (2020) Realizing an efficient IoMT-assisted patient diet recommendation system through machine learning model. IEEE Access 8:28462–28474
21. Iwendi C, Khan S, Anajemba JH, Mittal M, Alenezi M, Alazab M (2020) The use of ensemble models for multiple class and binary class classification for improving intrusion detection systems. Sensors 20(9):2559
22. Iwendi C, Moqurrab SA, Anjum A, Khan S, Mohan S, Srivastava G (2020) N-Sanitization: A semantic privacy-preserving framework for unstructured medical datasets. Comput Commun 161:160–171
23. Kerr OA, Tidman MJ, Walker JJ, Aldridge RD, Benton EC (2010) The profile of dermatological problems in primary care. Clin Exp Dermatol: Exp Dermatol 35(4):380–383
24. Li N, Zhao X, Yang Y, Zou X (2016) Objects classification by learning-based visual saliency model and convolutional neural network. Comput Intell Neurosci 2016. https://doi.org/10.1155/2016/7942501
25. Ramlakhan K, Shang Y (2011) A mobile automated skin lesion classification system. In: 2011 IEEE 23rd International Conference on Tools with Artificial Intelligence, pp 138–141
26. Ratul MAR, Mozaffari MH, Lee W, Parimbelli E (2020) Skin lesions classification using deep learning based on dilated convolution. BioRxiv, 860700
27. Rezvantalab A, Safigholi H, Karimijeshni S (2018) Dermatologist level dermoscopy skin cancer classification using different deep learning convolutional neural networks algorithms. arXiv preprint, arXiv:1810.10348
28. Ruiz D, Berenguer V, Soriano A, SáNchez B (2011) A decision support system for the diagnosis of melanoma: A comparative approach. Expert Syst Appl 38(12):15217–15223
29. Setting the learning rate of your neural network. https://www.jeremyjordan.me/nn-learning-rate/. Accessed 18 Nov 2020
30. Siegel RL, Miller KD, Jemal A (2020) Cancer statistics, 2020. CA A Cancer J Clin 70:7–30. https://doi.org/10.3322/caac.21590
31. Sladojevic S, Arsenovic M, Anderla A, Culibrk D, Stefanovic D (2016) Deep neural networks based recognition of plant diseases by leaf image classification. Comput Intell Neurosci 2016
32. Vijayalakshmi MM (2019) Melanoma skin cancer detection using image processing and machine learning. Int J Trend Sci Res Dev 3(4):780–784
33. Wang JL, Ibrahim AK, Zhuang H, Ali AM, Li AY, Wu A (2018) A study on automatic detection of IDC breast cancer with Convolutional Neural Networks. In: 2018 International Conference on Computational Science and Computational Intelligence - CSCI'2018, pp 703–708
34. Zeiler MD, Fergus R (2014) Visualizing and understanding convolutional networks. In: European conference on computer vision, pp 818–833